



Fluorite

△ **Finely crystallized**, multiple grouping of interpenetrating green fluorite cubes

Representing one of the widest colour ranges of any mineral, fluorite's hues tend to be vibrant, with violet, green, and yellow the most common. Its colours are commonly found as zones of different colours within a single crystal, and these zones typically follow the contour of the crystal faces. Fluorite is easily cleaved (broken along its atomic planes) and so is faceted for collectors only. For this, the stone is carefully oriented to avoid its four cleavage planes, and then cut and polished slowly to avoid heat or vibration.

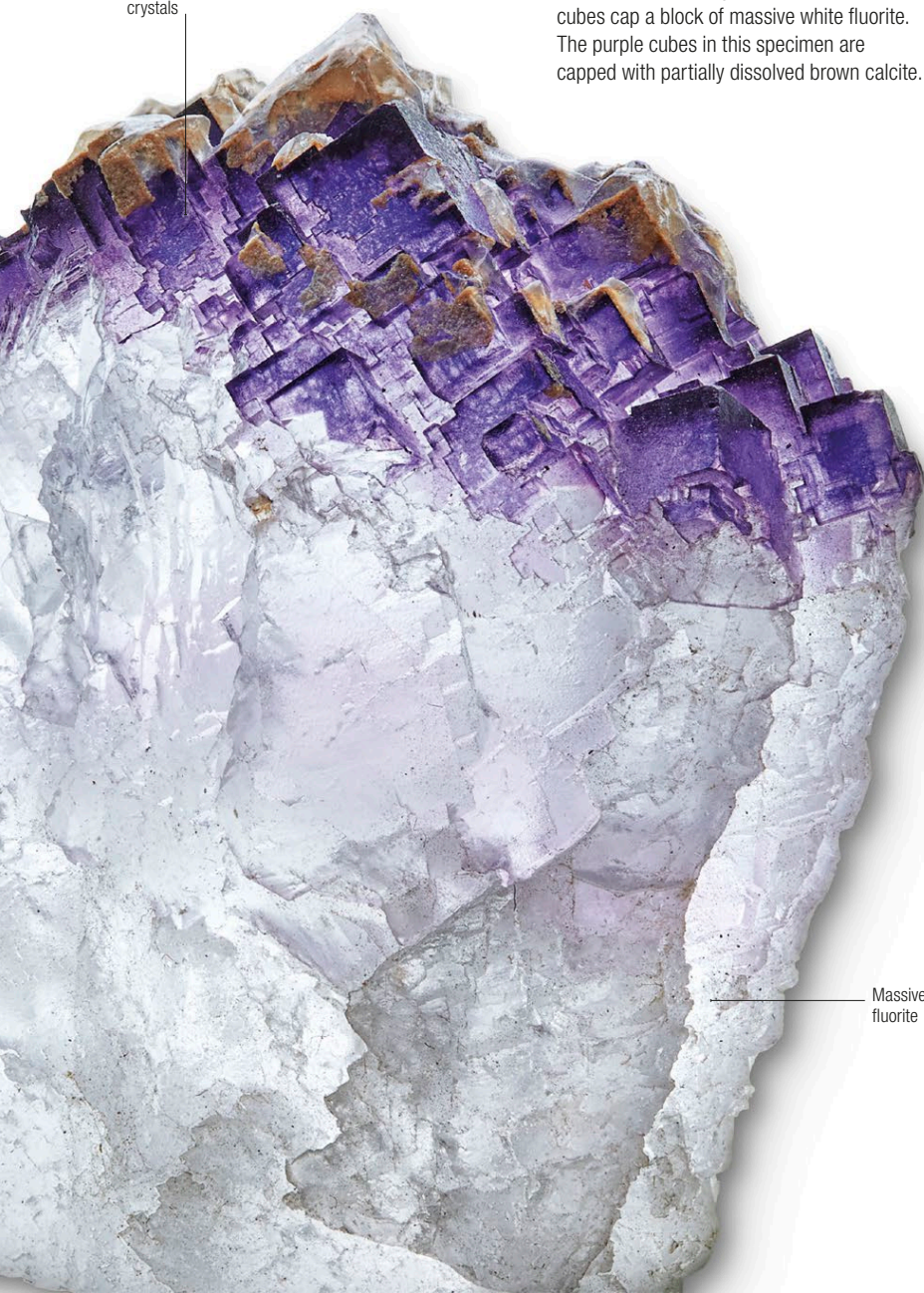
Specification

Chemical name Calcium fluoride | **Formula** CaF_2
Colours Colourless, blue, green, purple, orange | **Structure** Cubic
Hardness 4 | **SG** 3.0–3.3 | **RI** 1.43–1.44 | **Lustre** Vitreous
Streak White | **Locations** Canada, USA, Mexico, South Africa, China, Mongolia, Thailand, Peru, Europe, UK

Rough

Cubic crystals

Fluorite cubes | A group of purple fluorite cubes cap a block of massive white fluorite. The purple cubes in this specimen are capped with partially dissolved brown calcite.



Massive fluorite

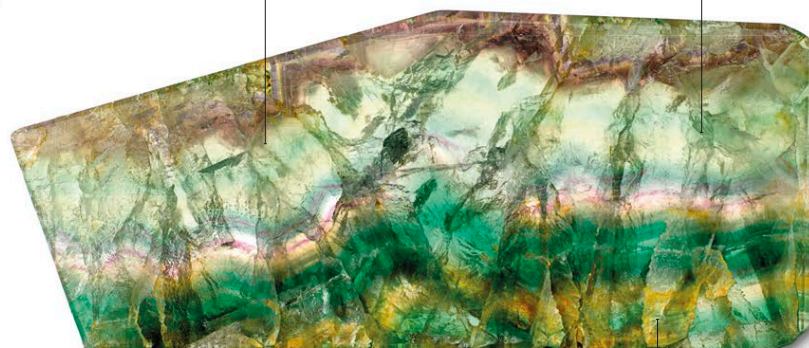
Colour layering



Blue John | This uncut piece of Blue John shows the layering of purple and yellow colours characteristic of this fluorite variety (see box, opposite).

Shadows of enclosed crystal

Colours vary from pale to dark



Fluorite slice | This thin, semi-transparent, slice of fluorite, around 5mm (¼in) in thickness, displays a variety of colours, as well as visible layering.

Colour zoning